

Pre-Test

Dedaring memory on the stack for a memory address pointing to radius

this constructor is constructing a SimpleCircle object

```
class SimpleCircle {
```

```
    double radius;
```

instance variable

```
    SimpleCircle() {  
        radius = 1;  
    }
```

```
    SimpleCircle(double newRadius) {  
        radius = newRadius;  
    }
```

```
    double getArea() {  
        return radius * radius * Math.PI;  
    }
```

```
    double getPerimeter() {  
        return 2 * radius * Math.PI;  
    }
```

```
    void setRadius(double newRadius) {  
        radius = newRadius;  
    }
```

getters and setters are examples of instance methods

class, constructor, overload, instance methods, instance variables,

private, public, encapsulation, getters, setters

object, reference variable, instance

Memory, stack, heap

— instance method

} getter methods

— setter method

```
public class TestSimpleCircle {
```

```
    /** Main method */
```

```
    public static void main(String[] args) {
```

this should be private! for encapsulation.

```
        SimpleCircle circle1 = new SimpleCircle();  
        System.out.println("The area of the circle of radius "  
            + circle1.radius + " is " + circle1.getArea());
```

```
        SimpleCircle circle2 = new SimpleCircle(25);  
        System.out.println("The area of the circle of radius "  
            + circle2.radius + " is " + circle2.getArea());
```

```
        SimpleCircle circle3 = new SimpleCircle(125);  
        System.out.println("The area of the circle of radius "  
            + circle3.radius + " is " + circle3.getArea());
```

reference variable

```
        circle2.radius = 100;  
        System.out.println("The area of the circle of radius "  
            + circle2.radius + " is " + circle2.getArea());
```

instance method using dot operator

class

new allocates memory on the heap